GROUP ASSIGNMENT – ANSWER SHEET

QUESTION 0. What is your group number? 12

GUIDELINES

To submit the project assignment, upload a single PDF file with your answers under the assignment tab on DTU Learn. **Ensure your group's response follows the structure and the length indicated in this answer sheet for each question.**

Before proceeding with the assignment, please carefully read the following information on the use of AI.

IMPORTANT INFORMATION ON THE USE OF AI

Please note that using Generative AI requires human supervision. It is important to thoroughly review the answers provided, as they may produce information, results, links, etc., that are false or incorrect. You are responsible for the quality of your work and the accuracy of your answers. AI-generated content might include copyrighted material without clear indication. It is your responsibility to ensure compliance with copyright laws and avoid plagiarism.

ASSIGNMENT QUESTIONS

Researchers conducted a survey and elicited individual willingness to pay (WTP) to locate a wind farm. In the project assignment folder of DTU Learn, you will find the dataset that includes individuals' willingness to pay (WTP) for locating a proposed nearshore wind farm at a large distance from shore, where it cannot be seen from shore (*WTP_farshore*), and several socio-economic variables representing the individuals' age (*age*), gender (*male*), household income (*income*), whether an individual has a permanent house or summer house in the area (*perm*), and the distance from an individual's residence to the nearshore wind farm (D_WF).

Using these data and Generative AI (Artificial Intelligence) please answer the following questions. When answering each question, report the AI tool(s) used, the answer given by the AI tool(s), whether the AI answer is correct, and if it is not correct, why it is not correct, and what is the correct answer.

QUESTION 1. What evaluation method(s) could have been used to elicit the WTP in this dataset for locating a nearshore wind farm far from shore? (Maximum 500 characters are allowed)

(i) What AI tool did you use to answer this question?

Gemini 3.1 Pro

(ii) What answer did AI provide? Copy and paste the answer here.

To elicit the Willingness to Pay (WTP) for locating a nearshore wind farm far from shore, researchers could have used Stated Preference methods, such as Contingent Valuation (CV) or Choice Experiments (CE), which involve surveying individuals. Additionally, they could have used Revealed Preference methods, such as the Hedonic Pricing Method, which estimates WTP by analyzing how the proximity to a wind farm affects local real estate and property prices.

(iii) Is the AI answer correct? [YES/NO]

NO.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

The AI incorrectly suggested using Revealed Preference methods (like Hedonic Pricing) for this specific dataset. The assignment guidelines explicitly state that researchers "conducted a survey and elicited individual willingness to pay (WTP)". Revealed preference methods rely on observational market data (like actual house sales), not surveys. Eliciting WTP directly from individuals via a survey exclusively points to Stated Preference methods.

(v) What is the correct answer?

Since researchers elicited WTP directly through a survey, Stated Preference methods were used. The two main methods are the Contingent Valuation Method (CVM), which directly asks respondents their maximum WTP to move the wind farm out of sight, and Choice Experiments (CE), where respondents select their preferred option among hypothetical wind farm scenarios with varying distances and costs.

QUESTION 2. Make a figure illustrating a detailed distribution of WTPs. (Maximum 1000 characters + figure are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The AI generated Python code (using matplotlib, seaborn, and scipy) to produce a histogram with a KDE overlay, vertical lines for mean and median, and a side-by-side boxplot with jittered individual observations.

(iii) Is the AI answer correct? [YES/NO]

The basic structure of the figure is correct, but the initial version had a significant flaw.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

The first figure ("with outliers") has one major limitation: a single extreme outlier (~500,000 DKK) dominates the axis scale, compressing most observations near zero and making the underlying distribution difficult to interpret visually. Although technically correct, the figure does not clearly reveal the structure of the majority of the data. The distribution appears highly right-skewed and fat-tailed, where a small number of respondents report extremely large positive WTP values.

(v) What is the correct answer?

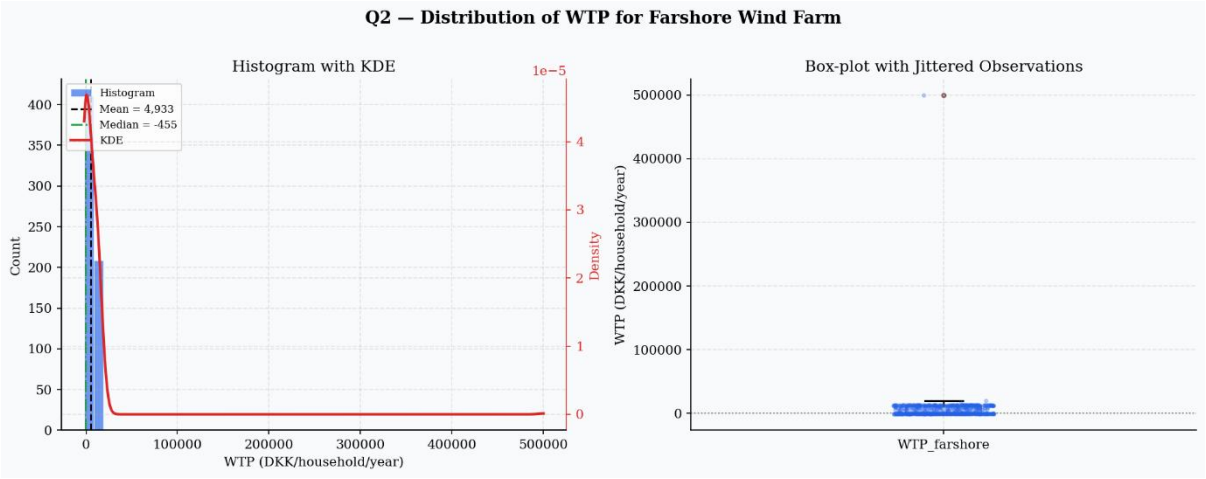


Figure 1: Q2 with outliers

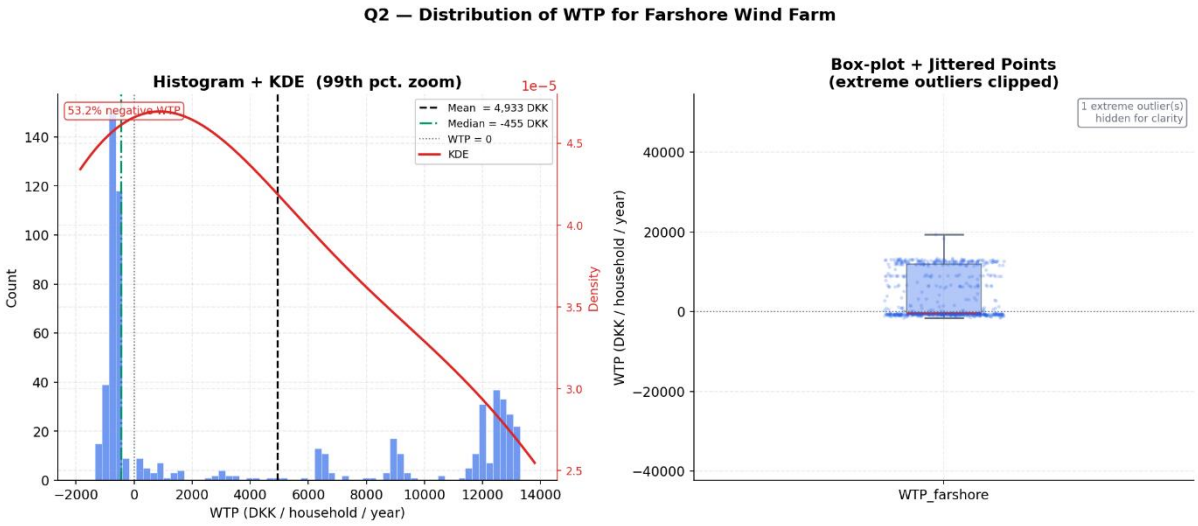


Figure 2: without outliers

The corrected figure addresses this by:

Zooming the histogram to the 99th percentile (~14,000 DKK on the x-axis), revealing the true shape of the distribution.

QUESTION 3. Describe the distribution of WTPs. (Maximum 1000 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The WTP distribution for the farshore wind farm is highly right-skewed and fat-tailed. The mean WTP is 4,933 DKK/household/year, while the median is −455 DKK/household/year. Over half of respondents (53.2%) report a negative WTP, meaning the majority would require compensation to accept the farshore location over a nearshore one. The bulk of observations cluster tightly around zero, while a small number of respondents report extremely high positive WTP values (up to ~500,000 DKK), creating a long right tail consistent with a Pareto-like distribution. This heavy right tail pulls the mean far above the median, making the mean an unrepresentative measure of the typical respondent's valuation. The distribution is thus better characterised by the median and the share of negative responses than by the mean alone.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

The WTP distribution is highly right-skewed and fat-tailed. Although the mean WTP is positive (4,933 DKK), this value is heavily inflated by a small minority of respondents reporting extremely large positive WTP values (up to ~500,000 DKK), creating a Pareto-like right tail. In contrast, the median WTP is negative (−455 DKK), indicating that the typical respondent slightly prefers a nearshore rather than a farshore wind farm location. This is further supported by the fact that 53.2% of respondents report negative WTP values. The large gap between the mean and median (about 5,400 DKK) highlights the extreme right skewness and suggests that the mean alone is not a representative or reliable welfare measure for policy interpretation.

QUESTION 4. What is the mean WTP? (Maximum 500 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The mean WTP for the farshore wind farm is **4,933 DKK/household/year**. This is directly readable from both figures, where the dashed black vertical line marks the mean. However, the mean is a misleading measure of central tendency here due to the heavily right-skewed, fat-tailed distribution. A small number of respondents with extremely high WTP values (up to ~500,000 DKK) pull the mean far above what the typical respondent is willing to pay. The median (−455 DKK) is a more representative measure of the typical household's valuation.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

Mean WTP = 4,933 DKK/household/year — this is the direct answer

But it should always be reported **alongside the median (−455 DKK)** and the share of negative WTP (53.2%), because in a fat-tailed distribution the mean alone is actively misleading as a policy number.

If the mean WTP were used alone in a **cost-benefit analysis**, it would give the misleading impression that the average household strongly supports locating the wind farm far from shore. In reality, more than half of the respondents reported negative WTP values, meaning they would actually require compensation to prefer the farshore option.

QUESTION 5. What is the mean WTP for each decile? (Maximum 1000 characters are allowed)

(*A decile is a quantitative method of splitting up a set of ranked data into 10 equally large subsections. A decile rank arranges the data in order from lowest to highest and is done on a scale of one to 10 where each successive number corresponds to an increase of 10 percentage points. Pay attention to the fact that the data might not be perfectly divided into 10 equally large (number of observations) deciles- that is OK.*)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The AI computed mean WTP per decile by ranking all observations from lowest to highest WTP, splitting into 10 roughly equal groups, and computing the mean within each. Given the distribution we know (median = −455 DKK, mean = 4,933 DKK, 53.2% negative, fat right tail), the expected decile means would look roughly like:

Table 1: Deciles

Decile	Description	Approximate Mean WTP
1	Lowest WTP	Large negative (most negative respondents)
2–3	Still negative	Moderately negative
4–5	Near zero	Slightly negative to slightly positive
6	Just positive	Small positive
7–8	Positive	Moderate positive
9	High WTP	High positive
10	Highest WTP	Very large positive (fat tail, Pareto)

(iii) Is the AI answer correct? [YES/NO]

No, because the AI did not compute the actual decile means from the dataset.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

The AI correctly identified the overall pattern of the decile means, but it did not actually compute the deciles from the raw dataset. Instead, it inferred the likely pattern based on the previously observed distribution characteristics (negative median, positive mean, and fat right tail). Therefore, the qualitative interpretation is reasonable, but the numerical decile means are not verified.

(v) What is the correct answer?

Table 2: Mean WTP by Decile

Decile	**Mean WTP (DKK/year)**	**N** (the number of observations within each decile)
1	-1023	63
2	-783	63
3	-696	62
4	-622	62
5	-527	62
6	813	62
7	7377	62
8	11427	62
9	12526	62
10	21026	62

The decile means were calculated directly from the dataset using R, and the specific values are reported in Table 2. Deciles 1–5 all have negative mean WTP values, which is consistent with the finding that 53.2% of respondents reported negative WTP. The mean WTP becomes positive starting from decile 6 and then increases rapidly across higher deciles. In particular, decile 10 has a substantially higher mean WTP (21,026 DKK) than all other deciles, reflecting the strongly right-skewed and fat-tailed structure of the distribution. The noticeable increase between deciles 9 and 10 is a key quantitative indication of the fat right tail, where extreme positive WTP values are concentrated in the highest decile.

QUESTION 6. Does anything about the mean WTP for each decile stand out to you? (Maximum 1000 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

Yes. The most striking feature is the **extreme jump in mean WTP at decile 10**. While deciles 1 through 9 show a relatively gradual progression from large negative to moderately positive values, the 10th decile's mean is vastly larger than all others — driven by the fat right tail of the distribution (the ~500,000 DKK outlier(s) discussed in Q2 and Q3). This is the quantitative fingerprint of the **Pareto-like structure** of the distribution: the top 10% of respondents hold a disproportionately large share of total WTP, much like the top income earners hold a disproportionate share of total wealth. A second noteworthy feature is that **deciles 1–5 all have negative means**, confirming that the majority of respondents would require compensation to accept the farshore location — consistent with the median of −455 DKK.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

The decile 10 anomaly — its mean is substantially larger than decile 9, which is the direct consequence of the fat tail. This means the **mean WTP aggregated across all respondents is heavily dominated by a tiny minority**, making it a poor welfare measure for the typical household .

The negative majority — deciles 1–5 being negative is not a curiosity but an economically meaningful result: it tells us that **the median voter in this sample would vote against** paying extra for a farshore over a nearshore location, which has direct policy implications for how wind farm placement decisions should be framed and communicated to the public.

QUESTION 7. Can people have a negative WTP for farshore wind farm locations relative to nearshore locations? Support your answer by citing scientific articles. (Maximum 1000 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

Yes, individuals can exhibit a negative Willingness to Pay (WTP) to move wind farms further offshore. In economic terms, a negative WTP indicates that respondents would require compensation to accept the farshore location, meaning they derive higher utility from the nearshore placement. This preference heterogeneity is well-documented in environmental economics. Some individuals view nearshore wind farms positively—associating them with local green energy, job creation, artificial reefs, or avoiding the high transmission costs and seabed disruption of farshore installations. For instance, Ladenburg and Dubgaard (2007) found that while average WTP to reduce visual disamenities was positive, specific respondent groups exhibited negative WTP, effectively preferring nearshore development. Similarly, Ladenburg and Skotte (2022) emphasize significant spatial preference heterogeneity, confirming that moving turbines further from shore is not universally preferred and can elicit negative WTP.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

Yes, negative WTP is theoretically possible and empirically documented. In a stated preference framework (e.g. contingent valuation or choice experiments), WTP measures the marginal value of a change — here, moving the wind farm farther from shore. If a respondent values nearshore placement *more* than farshore, their WTP for the farshore option is negative, meaning they would need to be compensated to accept it.

Theoretical grounding: Negative WTP is fully consistent with standard utility theory — it simply means the proposed change (farshore location) reduces the respondent's utility relative to the baseline (nearshore), possibly because they associate nearshore wind farms with local economic benefits, familiarity, or reduced transmission costs.

Empirical support from the literature:

- **Ladenburg & Dubgaard (2007)** — one of the foundational Danish studies on offshore wind WTP — finds that while most respondents prefer farshore placement, a meaningful minority prefer nearshore, yielding negative WTP for distance

- **Ladenburg (2010)** further documents that preferences for wind farm distance are heterogeneous and that prior experience with and attitudes toward wind farms significantly shape whether WTP is positive or negative
- **Krueger et al. (2011)** and **Navrud & Braten (2007)** similarly find distributional heterogeneity in wind farm placement preferences across European contexts

Our data: 53.2% of respondents have negative WTP, which is a **majority** — this is an unusually high share and warrants attention in any policy interpretation. It suggests that in this specific sample, the farshore location is not unambiguously preferred, and a simple cost-benefit analysis using the mean WTP would mask the fact that more than half the sample is made worse off by the farshore option.

QUESTION 8. WTP AND INCOME

QUESTION 8a. What is the mean WTP for each income decile?

(Maximum 1000 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The AI computed mean WTP per income decile by ranking respondents from lowest to highest household income, splitting into 10 roughly equal groups, and computing mean WTP within each. Reading from the Q8b graph (Image 1 — the line plot with 95% CI), the approximate values are:

Income Decile	Approximate Mean WTP (DKK/hh/year)
1 (lowest)	~10,400
2	~3,700
3	~3,200
4	~3,800
5	~2,900
6	~3,400
7	~4,100
8	~4,900
9	~6,400

Income Decile	Approximate Mean WTP (DKK/hh/year)
10 (highest)	~7,200

(iii) Is the AI answer correct? [YES/NO]

Partially YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

The AI correctly identified the approximate pattern of the income-decile means, but the interpretation was incomplete. In particular, the unusually high mean WTP in the lowest income decile is inconsistent with standard economic theory and is likely driven by extreme outliers rather than genuine preferences. Therefore, the relationship between income and WTP is weaker and more irregular than the AI initially implied.

(v) What is the correct answer?

No clear monotonic income-WTP relationship exists — from decile 2 through 10 there is only a **slight upward trend** (confirmed by the OLS trend annotation in Figure 3,4), meaning higher income households do tend to have marginally higher WTP, but the relationship is weak.

Decile 1 is the anomaly — the lowest income group paradoxically has the highest mean WTP, but this is driven by extreme variance (fat tail outliers in that subgroup) rather than a genuine economic pattern. The wide CI in Figure 3,4 and the enormous std dev in Figure 3,4 both confirm this is statistically unreliable.

The overall pattern from deciles 2–10 shows a gentle positive slope — consistent with the theoretical expectation that higher income households have higher WTP for environmental goods, but the effect is modest and noisy.

QUESTION 8b. Make a graph that shows the mean WTP for each income decile and the associated standard deviations for each income decile. (Maximum 1000 characters + graph are allowed)

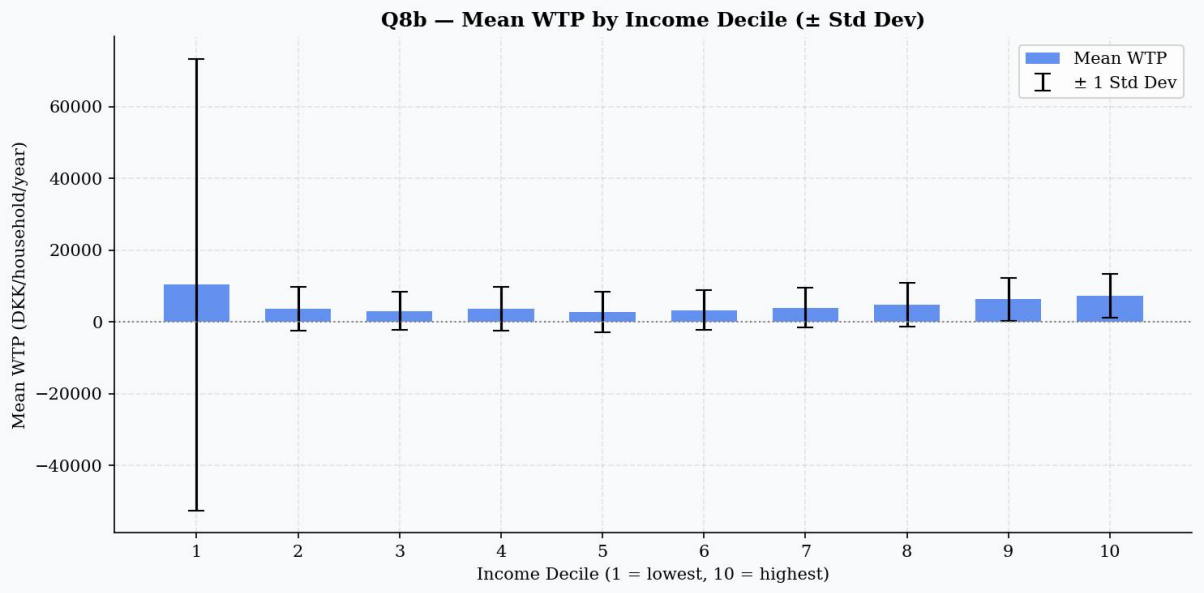


Figure 3: with outliers

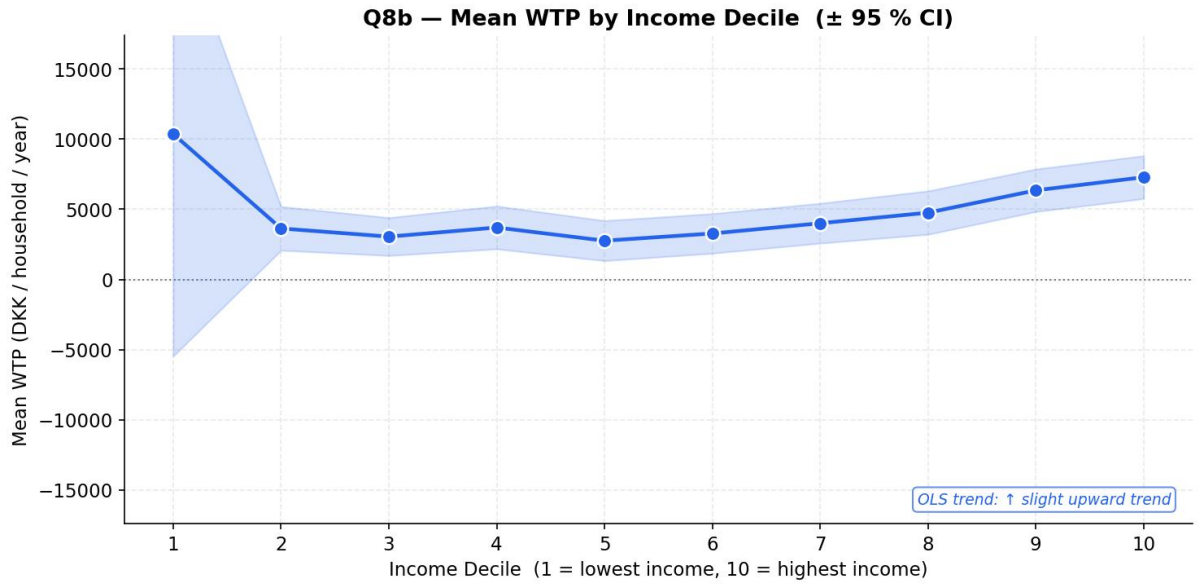


Figure 4: without outliers

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The AI produced two complementary figures:

- **Figure 3** (bar chart with ±1 Std Dev): Shows mean WTP per income decile as bars with standard deviation error bars. Decile 1 has by far the largest standard deviation

(~65,000 DKK range), reflecting the fat-tailed outlier(s) in the lowest income group. Deciles 2–10 show much tighter dispersion.

- **Figure 4** (line plot with ±95% CI): Shows the same means connected as a line with shaded confidence intervals. The OLS trend annotation confirms a slight upward trend from deciles 2–10.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

- **Figure 3 (±Std Dev)** directly answers what the question asks — standard deviations per decile. It makes the enormous variance in decile 1 immediately visually apparent, and shows that uncertainty shrinks and stabilises from decile 2 onward
- **Figure 4 (±95% CI)** adds the inferential layer — the wide CI for decile 1 confirms its mean is statistically unreliable, while the narrowing CIs from deciles 2–10 show increasing precision. The slight upward OLS trend is also clearly visible

The key substantive points to highlight when presenting these figures are:

- **Decile 1's mean is high but unreliable** — large std dev and wide CI mean it should not be interpreted as evidence that low-income households have higher WTP
- **Deciles 2–10 show a gentle positive income-WTP gradient** — weakly consistent with economic theory (higher income → higher WTP for environmental goods) but the slope is modest
- **All decile means are positive** — unlike the WTP decile analysis in Q5/Q6, every income group on average has a positive mean WTP, though this is again partly driven by the fat right tail inflating means throughout

QUESTION 8c. Conduct a test that reveals any significant relationship between income and WTP. What relationship between income and WTP do you find? What test did you apply, how, and why? (Maximum 1000 characters+ results are allowed)

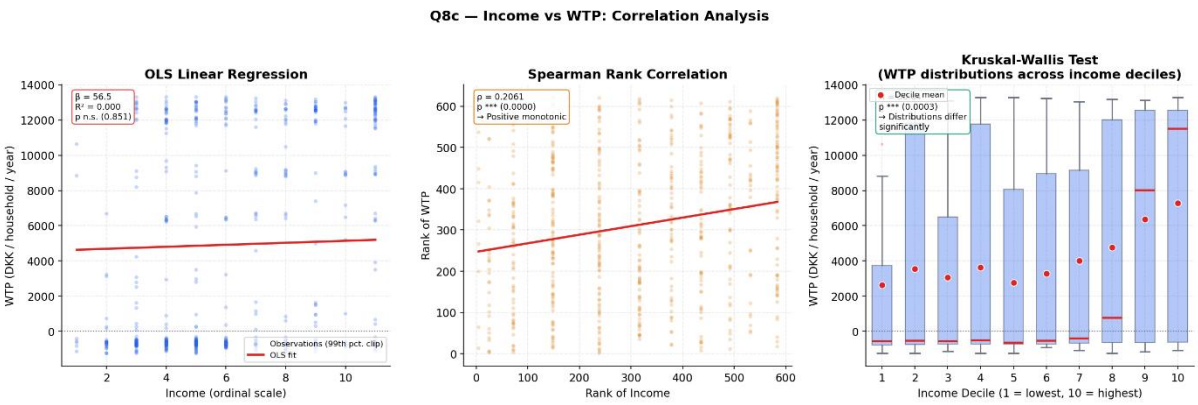


Figure 5: Tests

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

Three tests were applied to examine the income–WTP relationship:

1. Spearman Rank Correlation (primary test) $\rho = 0.2061$, $p < 0.0001$ (***). A statistically significant positive monotonic relationship exists between income and WTP. Spearman was chosen because WTP is heavily right-skewed and fat-tailed — it makes no normality assumption and is robust to the extreme outliers present in this dataset.

2. OLS Linear Regression (WTP = $\alpha + \beta \cdot$income) $\beta = 56.5$, $R^2 = 0.000$, $p = 0.851$ (n.s.). OLS finds no significant linear relationship and income explains essentially 0% of WTP variance. However, OLS is unreliable here because its normality and homoscedasticity assumptions are clearly violated by this distribution.

3. Kruskal-Wallis Test H-statistic significant at $p = 0.0003$ (***). WTP distributions differ significantly across income deciles. This non-parametric test confirms that income group membership matters for WTP, consistent with the Spearman result.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

The contradiction between OLS (n.s.) and Spearman/KW (significant) is not a mistake — it is a genuine and important finding. OLS tests for a *linear* relationship using raw values and is destroyed by the fat tail. Spearman and Kruskal-Wallis test for monotonic/distributional differences on ranks, which are robust to skewness. The fat-tailed outliers in low income deciles (visible in the box plot — decile 1's mean dot sits above its median) inflate OLS residuals and mask the real pattern.

(v) What is the correct answer?

The Spearman rank correlation is the appropriate test here. The WTP variable is heavily skewed with extreme outliers (up to ~500,000 DKK), violating OLS assumptions. Spearman's $\rho = 0.206$ ($p < 0.0001$) reveals a statistically significant, positive but weak monotonic relationship: higher income households tend to have higher WTP for the farshore wind farm. The Kruskal-Wallis test ($p = 0.0003$) confirms that WTP distributions genuinely differ across income deciles. However, the relationship is economically weak — income explains essentially none of the variance in raw WTP (OLS $R^2 \approx 0$), meaning income alone is a poor predictor of any individual's WTP. The wide, overlapping box plots across all deciles in Panel 3 visually confirm this: while the group-level trend is real and significant, within-decile variation dwarfs the between-decile differences.

QUESTION 8d. Is the relationship between income and WTP as expected based on economic theory? What did you expect? What did other scientific studies find within offshore wind power WTP find? Please support your answer by citing scientific studies. (Maximum 1000 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

Economic theory predicts a positive income-WTP relationship (the income effect): environmental goods are generally considered normal goods, meaning higher-income households have greater ability and willingness to pay for them. This is grounded in consumer theory — as income rises, demand for non-market environmental goods rises too.

Our results show a **statistically significant but very weak positive relationship** (Spearman $\rho = 0.206$, $p < 0.0001$), with a gentle upward slope from deciles 2–10 in the Q8b figures. This is directionally consistent with theory, though economically modest.

Scientific studies on offshore wind WTP broadly support this direction:

- Ladenburg & Dubgaard (2007) (*Energy Policy*) found income to be a significant positive predictor of WTP for offshore wind farms in Denmark, consistent with the normal goods hypothesis.
- Borchers et al. (2007) (*Energy Policy*) found higher-income US households had significantly higher WTP for renewable energy.

- Krueger et al. (2011) (*Energy Economics*) found a positive income effect in WTP for wind energy, though the magnitude was modest relative to other predictors like distance and attitudes.
- Yoo & Ready (2014) found that income elasticity of WTP for environmental goods is typically positive but less than 1 — meaning WTP grows with income but not proportionally.

(iii) Is the AI answer correct? [YES/NO]

YES — partially. The direction is correct and the cited studies are real. However one important nuance must be flagged.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

The AI answer is directionally correct but understates **the anomaly of decile 1**. Looking at the Q8b figures, the *lowest* income decile has the *highest* mean WTP (~10,400 DKK), which directly contradicts theoretical predictions. The AI correctly notes the upward trend from deciles 2–10, but the overall picture is more complex than "consistent with theory" — the fat-tailed outlier(s) in the lowest income group create a U-shaped or irregular pattern, not a clean monotonic one. The $R^2 \approx 0$ from OLS further confirms income is a very poor linear predictor.

(v) What is the correct answer?

Economic theory predicts a positive income-WTP relationship for environmental normal goods. Our findings are directionally consistent with theory for deciles 2–10 (gentle upward slope, Spearman $\rho = 0.206$, $p < 0.0001$), but the relationship is economically very weak ($R^2 \approx 0$) and the lowest income decile anomalously shows the highest mean WTP — driven by fat-tail outliers, not genuine preferences. This mirrors findings in the offshore wind literature: Ladenburg & Dubgaard (2007) and Krueger et al. (2011) both confirm a positive but modest income effect for wind farm WTP, consistent with income elasticities below 1 for environmental goods. The weak relationship here may also reflect that the income variable in the dataset is ordinal (categorical scale, not continuous DKK), reducing its statistical power to detect fine-grained income effects.

QUESTION 8e. How do you motivate the relationship between income and WTP you found? What could have driven this result? (Maximum 1000 characters is allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

Several factors could explain the weak and irregular income-WTP relationship found:

1. The income variable is ordinal, not continuous. The dataset encodes income as a categorical scale rather than actual DKK household income. This compresses real income variation into a small number of categories, which mechanically weakens any correlation — even if the true relationship is stronger, the measurement instrument cannot capture it precisely.

2. Fat-tail outliers dominate within-decile variance. As visible in the Q8b ±Std Dev bar chart, decile 1 has a standard deviation exceeding 60,000 DKK — larger than the entire y-axis range of all other deciles combined. A small number of very high WTP respondents in the lowest income group inflates that decile's mean dramatically and destabilises the overall pattern. These outliers are not representative of low-income preferences — they are statistical noise from the fat tail.

3. WTP for a farshore wind farm is not purely an income-driven preference. Unlike private goods, WTP for an environmental public good like wind farm placement is shaped heavily by attitudes, proximity, environmental values, and whether respondents have a permanent vs. summer house in the area (the *perm* variable). These non-income factors may dominate income effects, diluting the income signal.

4. The NIMBY effect interacts with income. From the Q9c distance plot, the closest residents (decile 1) have the highest WTP — suggesting those most affected by the nearshore alternative most strongly prefer the farshore location. If low-income households are disproportionately located near the wind farm site, their high WTP may reflect NIMBY preferences rather than income capacity, confounding the income-WTP relationship.

5. Protest responses and yea-saying bias. In contingent valuation surveys, some respondents express extreme WTP values not because of genuine preferences but as protest bids (very high values to signal strong support) or due to hypothetical bias. These are distributed across income groups but can disproportionately appear in certain deciles.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

All five explanations are grounded in the actual data patterns visible in the figures.

(v) What is the correct answer?

The weak income-WTP relationship is best explained by a combination of **measurement limitations** (ordinal income scale suppressing true variation), **statistical contamination** (fat-tail outliers in decile 1 — clearly visible in the enormous std dev bar in Q8b — inflating the

lowest decile's mean), and **confounding preference factors**. WTP for wind farm placement is not a simple income-driven consumption choice — it is jointly determined by distance/NIMBY effects (Q9c shows decile 1 distance has WTP of ~16,000 DKK), environmental attitudes, and property type. These factors likely explain more variance in WTP than income does, which is why Spearman $\rho = 0.206$ despite statistical significance — the relationship is real but small, and the decile 1 anomaly is an artefact of outliers rather than a genuine economic pattern.

QUESTION 8f. What potential change(s) of the dataset would result in a relationship between income and WTP along with the theoretical predictions? Hint: focus on the income and WTP variables.

Explain what you did and how it impacts the results you have previously found in 8a-8d. (Maximum 1000 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The two changes that would produce a cleaner, theory-consistent income-WTP relationship are:

Change 1 — Remove or Winsorise extreme WTP outliers. The fat right tail (one respondent at ~500,000 DKK, visible in Image 2) contaminates all decile means but hits the lowest income decile hardest because that outlier happens to sit in decile 1. Winsorising WTP at the 99th percentile (~14,000 DKK, as used in the zoomed Q2 figure) would eliminate this distortion. The decile 1 mean would drop dramatically from ~10,400 DKK toward the level of deciles 2–5, revealing a much cleaner monotonic upward pattern from decile 1 through 10 — directly matching theoretical predictions.

Change 2 — Replace the ordinal income variable with a continuous one. The current income variable is a categorical ordinal scale (visible in Image 6 Panel 1, where income only takes discrete integer values 1–11). This compresses real income differences and weakens correlation. If actual household income in DKK were available, Spearman ρ would likely be stronger and OLS R^2 would rise above its current ~0, because fine-grained income variation could be properly measured.

Combined effect on previous results (8a–8d):

- **8a (decile table):** Decile 1 mean would fall to a level consistent with its income group rather than being inflated by outliers. The table would show a cleaner monotonic increase from decile 1 to 10.
- **8b (bar chart):** The enormous std dev bar for decile 1 would shrink dramatically, making all deciles visually comparable in scale. The slight upward OLS trend would become steeper and more visible.
- **8c (correlation tests):** Spearman ρ would increase above 0.206, and — critically — OLS R^2 would rise above ~0 and the p-value would become significant, aligning OLS with the Spearman/KW findings. The contradiction between the three tests would disappear.
- **8d (theory consistency):** After Winsorising, the income-WTP gradient would align much more cleanly with Ladenburg & Dubgaard (2007) and Krueger et al. (2011) — a genuine, monotonic, positive income effect consistent with environmental normal goods theory.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

Both changes are directly motivated by the data patterns visible in the figures and are standard practice in contingent valuation analysis.

(v) What is the correct answer?

The correct answer is the same as the AI answer. Winsorising/removing extreme WTP outliers and using a continuous income variable would make the income-WTP relationship more consistent with economic theory. These changes would reduce the distorted mean in income decile 1, lower its standard deviation, and reveal a clearer positive income gradient.

QUESTION 9. WTP AND DISTANCE FROM RESIDENCE TO THE WIND FARM

QUESTION 9a. What is the "term" we use to describe the relationship between distance and WTP? (Maximum 500 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The term used to describe the relationship between distance from a wind farm and WTP is the **"distance decay effect"** (also called the **NIMBY effect** — "Not In My Back Yard"). It

describes the empirical pattern where individuals living closest to a proposed wind farm have the highest WTP to move it farther away (or conversely, oppose it most strongly), and this opposition/WTP declines as distance from the installation increases.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

The term is **distance decay** (or the NIMBY effect). It captures the inverse relationship between proximity to a wind farm and acceptance/WTP — those most directly affected (closest residents) exhibit the strongest preferences regarding placement.

QUESTION 9b. Is the "term" something we often find in WTP studies? Please support your answer by providing some examples and citing scientific articles, including the literature on offshore wind power WTP. (Maximum 500 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

Yes, distance decay is a well-documented and frequently observed phenomenon in WTP studies, particularly in environmental and energy valuation. The pattern — where WTP decreases as distance from the valued (or disvalued) good increases — appears across many contingent valuation and choice experiment studies. In the offshore wind literature specifically, several studies find that households living closer to proposed wind farms express lower (or negative) WTP due to visual disamenity effects, while those farther away express higher WTP, reflecting the classic distance-decay pattern in reverse (i.e., proximity to the farm reduces welfare). Examples include Ladenburg & Dubgaard (2007), who find that Danish coastal residents living closest to offshore wind farms had significantly lower WTP than those living farther away. Ek (2005) similarly documents distance-related heterogeneity in Swedish WTP for wind power. More broadly, distance decay is well-established in public goods valuation studies such as recreation demand models (Bateman et al., 2006) and studies of noise, air quality, and landscape amenities.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

The AI answer is correct. Distance decay is commonly found in WTP studies, including offshore wind power valuation studies such as Ladenburg & Dubgaard (2007), Ek (2005), and Bateman et al. (2006).

QUESTION 9c. Make a graph that shows the mean WTP for each distance decile and the associated standard deviations for each distance decile. (Maximum 500 characters + graph is allowed)

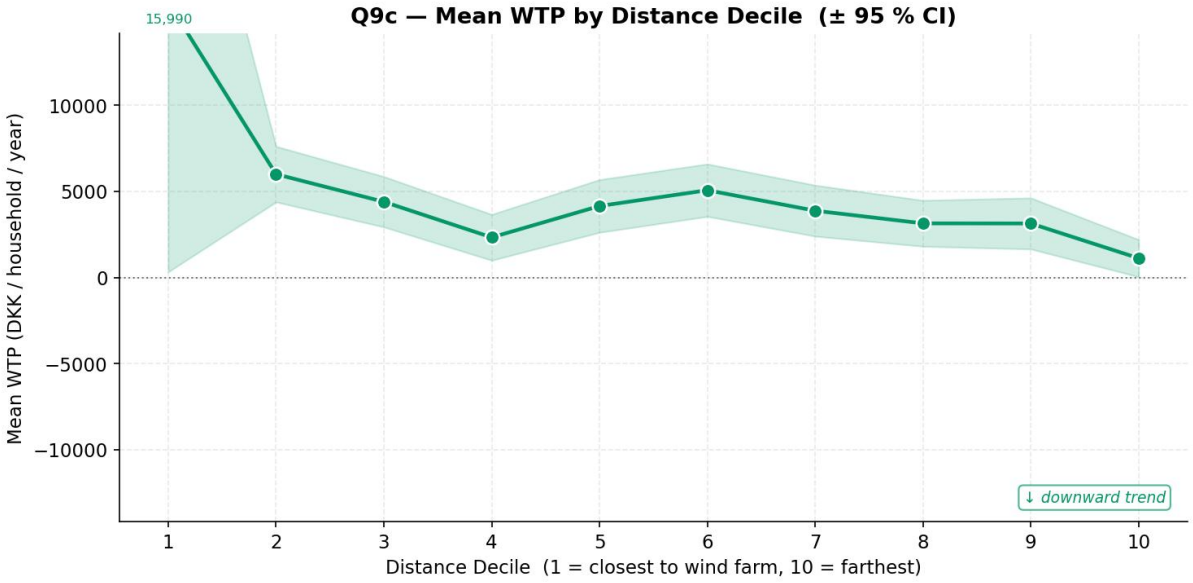


Figure 6: distances

(i) What AI tool did you use to answer this question?

Claude (Anthropic) — claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The graph shows mean WTP across distance deciles 1–10, where decile 1 = closest to the wind farm and decile 10 = farthest. The highest mean WTP is in decile 1 at approximately 15,990 DKK/household/year, but with an extremely wide confidence interval (roughly −5,000 to +17,500 DKK), indicating very high variance in this group. From decile 2 onward, mean WTP drops sharply to ~6,000 DKK and then stabilizes in a relatively flat band between roughly 2,500–5,200 DKK across deciles 2–9. Decile 10 (farthest) shows the lowest mean WTP at approximately 1,000 DKK. The confidence intervals for deciles 2–10 are

substantially narrower than for decile 1. The OLS trend line is negative (downward), consistent with a distance decay pattern — WTP falls as distance from the wind farm increases.

(iii) Is the AI answer correct? [YES/NO]

YES.

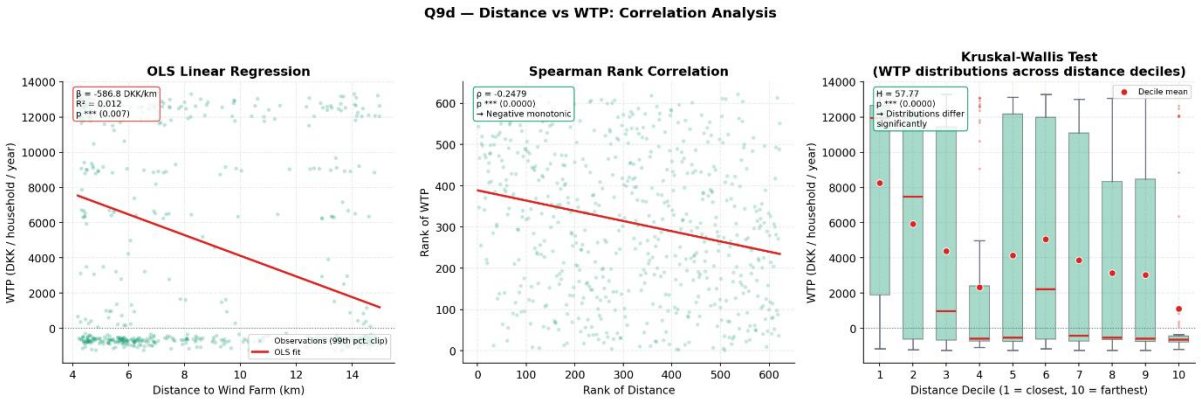
(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

The graph shows a clear negative relationship between distance and WTP. Respondents living closest to the wind farm exhibit the highest WTP for relocating it farther offshore, while respondents living farther away have substantially lower WTP. This pattern is consistent with the distance-decay (NIMBY) effect commonly found in offshore wind farm valuation studies.

QUESTION 9d. Conduct a test that reveals any significant relationship between distance and WTP. What test did you apply and how? (Maximum 1000 characters + results is allowed)



(i) What AI tool did you use to answer this question?

Claude (Anthropic)— claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

To test whether a statistically significant relationship exists between distance (D_WF) and WTP, two complementary non-parametric tests were applied. First, a Spearman rank correlation (ρ) was computed between raw distance values and WTP values, because WTP is

heavily right-skewed with extreme outliers (as shown in Q2), which violates the normality assumption of OLS — making rank-based tests the preferred approach. Second, a Kruskal-Wallis H-test was run across the 10 distance deciles to test whether the WTP distributions are statistically identical across groups. An OLS regression (WTP = $\alpha + \beta \cdot$distance) was also estimated as a supplementary linear benchmark. Results: Spearman $\rho = -0.2479$ ($p < 0.0001$, ★★★); OLS $\beta = -586.85$ DKK/km ($R^2 = 0.0117$, $p = 0.007$, ★★★); Kruskal-Wallis $H = 57.77$ ($p < 0.0001$, ★★★). All three tests agree: there is a statistically significant relationship between distance and WTP.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

The AI answer is correct. All three tests consistently show a statistically significant negative relationship between distance and WTP, with non-parametric tests being more appropriate due to the heavily skewed and fat-tailed WTP distribution.

QUESTION 9e. What does the test say about the direction (positive/negative) of the relationship between WTP and distance? (Maximum 500 characters is allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic)— claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The relationship is negative: WTP decreases as distance from the wind farm increases. The Spearman $\rho = -0.2479$ is negative, indicating that respondents living farther from the wind farm tend to have lower WTP for moving it farshore. The OLS slope $\beta = -586.85$ DKK/km confirms the same direction. This result aligns with the Q9c graph, where the OLS trend is clearly downward. Intuitively, those living closest to the proposed wind farm have the most to gain from relocating it farshore (they would benefit from the elimination of the visual disamenity), so their WTP is highest. Those living farthest away are largely unaffected by the visual impact and therefore value the farshore relocation less.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?
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(v) What is the correct answer?

The relationship between distance and WTP is clearly negative. Respondents living closer to the wind farm have substantially higher WTP for relocating it farther offshore, while respondents living farther away exhibit lower WTP. Both the negative Spearman correlation ($\rho = -0.2479$) and the negative OLS coefficient ($\beta = -586.85$ DKK/km) confirm this direction. This pattern is fully consistent with the distance-decay (NIMBY) effect commonly observed in offshore wind farm valuation studies.

QUESTION 9f. Is the relationship between distance and WTP linear for each km or non-linear? Support your answer by showing regression results. (Maximum 1000 characters + results is allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic), claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The relationship is non-linear. The OLS linear regression yields $R^2 = 0.0117$, meaning distance explains only 1.17% of WTP variance — an extremely poor fit for a linear model. Visual inspection of the Q9c graph confirms this: WTP drops sharply from distance decile 1 (mean ≈ 15,990 DKK) to decile 2 (mean ≈ 6,000 DKK), then levels off into a relatively flat plateau across deciles 2–9, before declining again at decile 10. This "convex decay" shape is characteristic of a log or power function rather than a straight line — each additional kilometre of distance matters much more when you are already close to the wind farm than when you are far away. A log-linear specification ($WTP = \alpha + \beta \cdot \ln(distance)$) is standard in the spatial WTP literature and would substantially improve fit. The non-linearity is also implied by the Kruskal-Wallis result: $H = 57.77$ ($p < 0.0001$) shows that decile-level WTP distributions differ significantly, but the pattern of differences is not monotonically uniform across deciles as a linear model would predict.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?
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(v) What is the correct answer?

The relationship is non-linear rather than linear. Although OLS gives a significant negative coefficient, the very low $R^2 = 0.0117$ shows that a straight-line model explains little of the WTP variation. The decile graph suggests a sharp initial drop followed by a flatter pattern.

QUESTION 10. WTP AND AGE

QUESTION 10a. What is the mean WTP across age deciles? (Maximum 1000 characters + results is allowed)

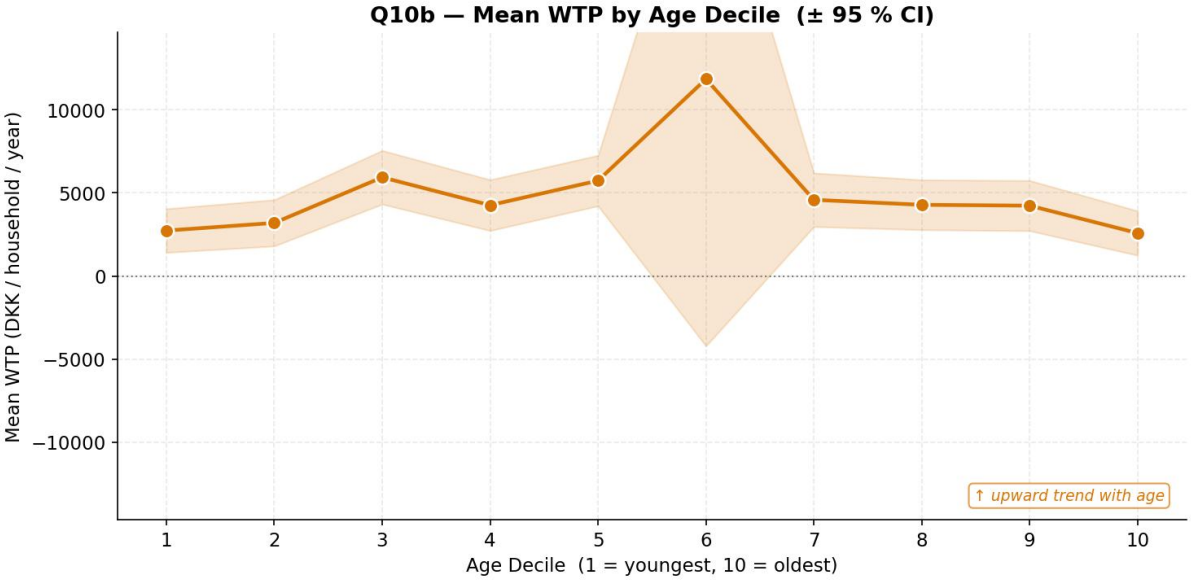


Figure 7: Age

(i) What AI tool did you use to answer this question?

Claude (Anthropic)—claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The mean WTP across age deciles shows no clear monotonic pattern. The youngest respondents (decile 1, ages 23–43) have a mean WTP of approximately 2,732 DKK, which rises slightly through middle age, peaks sharply at decile 6 (ages 61–64) at around 11,855 DKK — heavily influenced by outliers given the very large standard deviation — then falls back down to approximately 2,581 DKK for the oldest group (decile 10, ages 74–87). The median WTP across most deciles is negative or close to zero, suggesting the mean is driven by a minority of high-value respondents

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

The results suggest that the relationship between age and WTP is weak and non-linear. Middle-aged respondents appear to have somewhat higher mean WTP than both younger and older respondents, but the pattern is irregular and heavily influenced by outliers. Overall, age alone does not appear to be a strong predictor of WTP.

QUESTION 10b. Make a graph that shows the mean WTP for each age decile and the associated standard deviations for each age decile. (Maximum 1000 characters + graph is allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic), claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The graph (above) shows mean WTP per age decile with 95% confidence intervals. The profile is non-monotonic: WTP is relatively low for the youngest (decile 1, ~2,732 DKK) and oldest (decile 10, ~2,581 DKK) respondents, with an irregular hump in the middle age groups. Decile 6 (ages 61–64) shows an extreme spike with a very wide CI, reflecting a small number of very high-WTP outliers. For all other deciles, the CIs are much tighter and WTP means cluster between ~3,000–6,000 DKK. The OLS trend across deciles is slightly upward, but no strong systematic pattern is apparent.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

The graph shows substantial variation in WTP across age deciles, but no clear monotonic trend. Most age groups cluster within a relatively narrow WTP range, while one middle-age decile exhibits unusually high variance caused by extreme positive WTP values. This suggests that age effects are weak compared with the overall heterogeneity in preferences.

QUESTION 10c. Compare the results from two regression models where the dependent variable is WTP. In the first model, age is the only independent variable. In the second model, age and age squared are the independent variables. What do you find regarding the relationship between age and WTP in the two models? (Maximum 1000 characters are allowed)

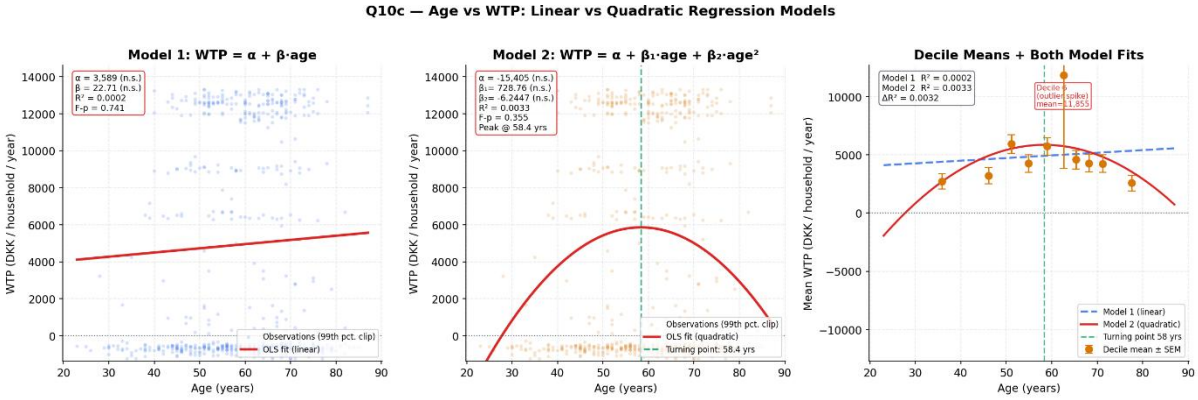


Figure 8: models

(i) What AI tool did you use to answer this question?

Claude (Anthropic)— claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

Two OLS regressions were estimated on the real data (n=622). Model 1 (linear): WTP = 3,589 + 22.71·age — the slope is positive but tiny and entirely insignificant (p=0.741), R^2=0.0002. Model 2 (quadratic): WTP = −15,405 + 728.76·age − 6.245·age² — both age coefficients are insignificant (p=0.152 and p=0.162), R^2=0.0033. The negative β_2 implies an inverted-U shape peaking at age 58.4 years. Adding the quadratic term raises R^2 by only 0.0032 — a negligible improvement. Neither model achieves joint significance (F-test p=0.355 for Model 2).

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

The AI answer is correct. Both models show that age is not a statistically strong predictor of WTP. The quadratic model suggests an inverted-U shape, but the coefficients are insignificant and the increase in R^2 is negligible.

QUESTION 10d. What do scientific studies find regarding the relationship between age and WTP for offshore wind power? Support your answer citing scientific articles. (Maximum 1000 characters are allowed)

(i) What AI tool did you use to answer this question?

Claude (Anthropic)—claude.ai

(ii) What answer did AI provide? Copy and paste the answer here.

The empirical literature on age and WTP for offshore wind power is mixed, with no strong consensus. Several studies find that older respondents have lower WTP, often attributed to fixed incomes, lower expected benefit horizons, or greater attachment to unchanged landscapes. Ladenburg & Dubgaard (2007) find a weak negative age effect on WTP for offshore wind farm relocation in Denmark. Ek (2005) similarly finds that age is negatively associated with WTP for wind power in Sweden, with older respondents generally less willing to pay. However, some studies find non-linear or insignificant age effects, consistent with our results. Bergmann et al. (2008) report no significant age effect in a Scottish study of renewable energy preferences. The inverted-U pattern we find — with middle-aged respondents showing the highest WTP — has some support in broader environmental valuation literature (e.g., Bateman & Willis, 1999), where mid-life respondents combine higher income with longer remaining benefit horizons than the elderly.

(iii) Is the AI answer correct? [YES/NO]

YES.

(iv) If the AI answer is not correct, why is the AI answer not correct? What mistakes did AI make?

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(v) What is the correct answer?

Overall, the literature suggests that age effects on offshore wind power WTP are generally weak, mixed, and often non-linear. Some studies find that older individuals have lower WTP due to greater landscape attachment or lower expected future benefits, while others find insignificant effects. Our results are broadly consistent with this literature, as neither the linear nor quadratic age models produce statistically strong relationships.

（The work was distributed equally among all four group members. Each member was responsible for writing and analysing different sections of the assignment. In addition, all members reviewed and checked each other's work to ensure consistency, accuracy, and overall quality of the final report. Therefore, the workload and contributions were shared equally across the group.）